1 Solution — GIAM 1.1 Problem 10

1.1 Problem

The *conjugate* of a complex number is denoted with a superscript star and is formed by negating the imaginary part. Thus if $z = 3 + 4i$ then $z^* = 3 - 4i$. Give an argument as to why the product of a complex number and its conjugate is a real quantity. (I.e. the imaginary part of $z \cdot z^*$ is necessarily 0, no matter what complex number is used for z.)

1.2 The Proof — Direct FOIL

Let $z = a + bi$ with $a, b \in \mathbb{R}$. Then $z^* = a - bi$.

Multiply via FOIL (or, recognising the **difference-of-squares** pattern with "$A = a$, $B = bi$"):

$$z \cdot z^* \;=\; (a + bi)(a - bi) \;=\; a^2 \;-\; (bi)^2 \;=\; a^2 \;-\; b^2\,i^2 \;=\; a^2 \;+\; b^2.$$

Since $a, b \in \mathbb{R}$, we have $a^2, b^2 \in \mathbb{R}$, and so $a^2 + b^2 \in \mathbb{R}$. In fact $a^2, b^2 \geq 0$, so

$$z \cdot z^* \;=\; a^2 + b^2 \;\in\; \mathbb{R}_{\geq 0}.$$

The imaginary part of $z \cdot z^*$ is 0 — for *any* choice of $z \in \mathbb{C}$. ■

1.3 Worked Example

For the textbook's $z = 3 + 4i$, $z^* = 3 - 4i$:

$$z \cdot z^* \;=\; (3 + 4i)(3 - 4i) \;=\; 3^2 + 4^2 \;=\; 9 + 16 \;=\; 25.$$

A real number — and in fact a perfect square, 5^2. (This is no coincidence: $|z| = 5$.)

1.4 Geometric Picture — Reflection Across the Real Axis

$z = a + bi$ and $z^* = a - bi$ are **mirror images of each other across the real axis** in the complex plane. Their product sits on the real axis itself, at a distance equal to the squared modulus $|z|^2 = a^2 + b^2$.

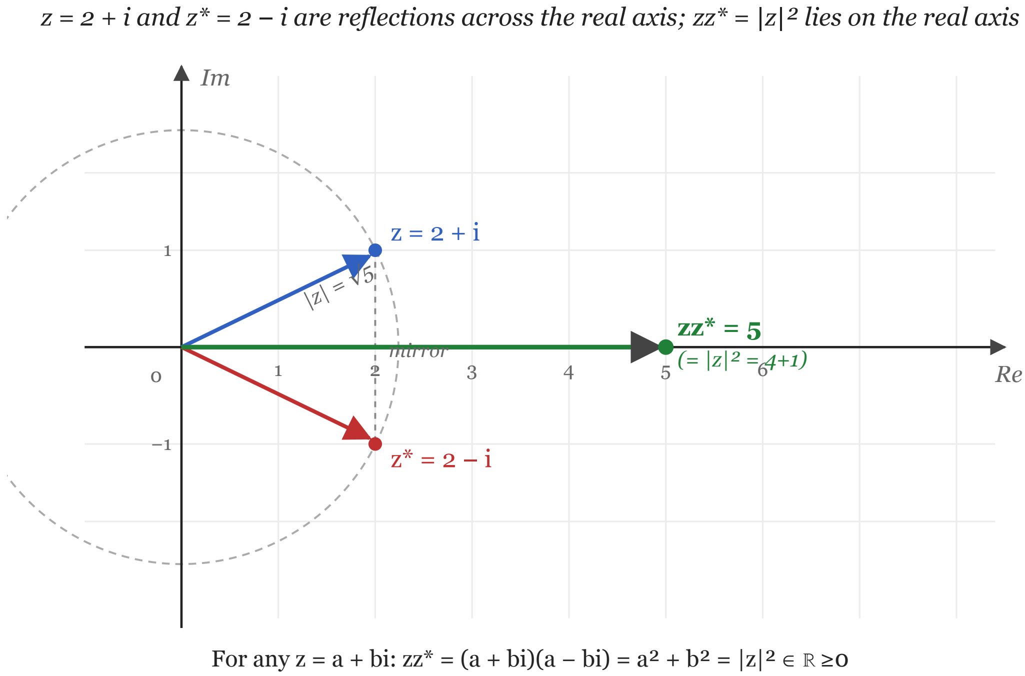

Figure 1.1: For z = 2 + i the conjugate z* = 2 − i is the mirror image across the real axis. The two arrows have the same length |z| = √5; their product zz* = 5 sits on the positive real axis at distance |z|² = 4 + 1.

1.5 Polar-Form Proof

The fastest argument is polar. Write $z = r\,e^{i\theta}$ with $r = |z|$ and $\theta = \arg z$. Then

$$z^* \;=\; r\,e^{-i\theta}.$$

(The conjugate has the same modulus and the opposite argument — reflection across the real axis in polar terms.) So

$$z \cdot z^* \;=\; r\,e^{i\theta} \cdot r\,e^{-i\theta} \;=\; r^2\,e^{i(\theta-\theta)} \;=\; r^2\,e^0 \;=\; r^2 \;=\; |z|^2.$$

The arguments cancel; only the (squared) modulus survives. So zz^* is real and non-negative. ■

1.6 The Identity $z \cdot z^* = |z|^2$

Both proofs above are special cases of the same identity:

$$z \cdot z^* \;=\; |z|^2 \;\in\; \mathbb{R}_{\geq 0}.$$

This identity is so important that it's often taken as the *definition* of $|z|$ for z complex: $|z| := \sqrt{z\,z^*}$. For real numbers the formula reduces to $\sqrt{x \cdot x} = |x|$ — the usual real absolute value.

1.7 Remark — Why "Necessarily 0" — A Symmetry Argument

There is an *even* shorter argument, just from the algebra of conjugation. The conjugation map $z \mapsto z^*$ satisfies these properties:

- $(z + w)^* = z^* + w^*$ (additivity)
- $(z \cdot w)^* = z^* \cdot w^*$ (multiplicativity)
- $(z^*)^* = z$ (involution — conjugate twice and you're back)
- $z^* = z$ iff $z \in \mathbb{R}$ (fixed points of conjugation are exactly the reals)

Apply property 4 backwards: to show $zz^* \in \mathbb{R}$, it suffices to show $(zz^*)^* = zz^*$. Compute using properties 2 and 3:

$$(zz^*)^* \;=\; z^* \cdot (z^*)^* \;=\; z^* \cdot z \;=\; z \cdot z^*.$$

So zz^* is *fixed* by conjugation — i.e. it equals its own conjugate — i.e. it is real. ■

This argument is *coordinate-free*: it never opens z as "$a + bi$". It uses only the four algebraic properties of $*$, which characterise it abstractly.

1.8 Remark — The Identity Powers Complex Division

The most practical use of $zz^* = |z|^2$ is in *dividing* complex numbers. To compute w/z, multiply numerator and denominator by z^*:

$$\frac{w}{z} \;=\; \frac{w \cdot z^*}{z \cdot z^*} \;=\; \frac{w \cdot z^*}{|z|^2}.$$

The denominator becomes a *real* number ($|z|^2$), and dividing a complex number by a real is trivial — just divide the real and imaginary parts separately. So:

$$\frac{1}{3+4i} = \frac{3-4i}{(3+4i)(3-4i)} = \frac{3-4i}{25} = \frac{3}{25} - \frac{4}{25}i.$$

Without the identity $zz^* \in \mathbb{R}$, complex division would be a much harder construction.

1.9 Remark — The Stronger Conclusion: $zz^* \geq 0$, with Equality Iff $z = 0$

The proof actually gives more than "zz^* is real": it gives $zz^* = a^2 + b^2 \geq 0$, with **equality iff** $a = b = 0$ (i.e. iff $z = 0$).

This is exactly what's needed to make $|z| = \sqrt{zz^*}$ a *norm*: a real-valued function on $\mathbb{C}$ that is positive except at 0, where it vanishes. It also makes the squared-distance function $d(z, w) = |z - w|^2 = (z - w)(z^* - w^*)$ behave like a proper metric — non-negative, zero iff $z = w$, symmetric, triangle inequality.

So the modest observation "zz^* is real" is what underwrites the *entire metric geometry* of the complex plane.

1.10 Remark — Other "Obvious" Sums via the Same Trick

The identity $zz^* = a^2 + b^2$ identifies the *sum of two squares* as the product of two specific factors. This is why **Gaussian integers** $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ provide a clean factorisation theory for sums of squares:

- $5 = (2 + i)(2 - i)$ as a Gaussian-integer product → so 5 is a sum of two squares: $2^2 + 1^2$.
- $13 = (3 + 2i)(3 - 2i) \rightarrow 13 = 3^2 + 2^2$.
- $25 = (3 + 4i)(3 - 4i) \rightarrow 25 = 3^2 + 4^2$ (and also $5^2 + 0^2$).

This is the start of **Fermat's two-squares theorem**: a prime p is a sum of two squares iff $p = 2$ or $p \equiv 1 \pmod 4$. The proof rests on Gaussian-integer arithmetic, which in turn rests on $zz^* \in \mathbb{R}_{\geq 0}$.

So this "trivial" exercise sits at the doorway of a rich theory.

1.11 Remark — Conjugation Reverses Direction in Polar Form

The polar-form proof revealed the *geometric* essence of conjugation: z^* has the **same modulus** as z and the **opposite argument**. Equivalently:

- $$|z^*| \;=\; |z|$$
- $$\arg z^* \;=\; -\arg z$$

So conjugation is the geometric operation "reflect through the real axis." Reflecting twice gives back the identity — matching the algebraic involution property $(z^*)^* = z$. Reflecting through the imaginary axis is a *different* operation (negation: $z \mapsto -z$), and reflecting through the origin is yet another ($z \mapsto -z$ again). Conjugation is uniquely tied to the *real axis* — which is exactly why zz^* lands on it.